SARATH RADHAKRISHNAN

Postdoctoral Researcher

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https://github.com/sarath4u/

PROFESSIONAL SUMMARY

Computational engineer with 7 years of experience in CFD, Large-Eddy Simulation (LES), non-linear flow physics, and scientific computing. Developed a data-driven wall model for LES of turbulent wall-bounded flows during doctoral research, combining turbulence modeling with machine-learning methods. Brings aerospace propulsion experience from SCRAMJET systems engineering at ISRO to large-scale CFD and scientific computing research. Contributed to reduced-order modeling by generating and analysing high-fidelity urban flow simulation datasets. Mentored graduate students and delivered science outreach.

EDUCATION

Universitat Politecnica de Catalunya *Sep 2018 – Jun 2024*

PhD *Barcelona, Spain*

Thesis: Non-Equilibrium Wall-modeling in Large Eddy Simulation of High-speed Transitional Flows. Advisors: Oriol Lehmkuhl and Daniel Mira.

Ecole Centrale de Nantes *Sep 2015 – Aug 2017*

Master of Engineering in Computational Mechanics *France*

Thesis: Modelling of Uterine Synchronisation by a Hydro-Mechanical and Electrical Activation Mechanism — Mechanical Modelling of Uterus. Advanced coursework in continuum mechanics, numerical methods, finite element methods, and computational modelling. Internship project funded by ANSYS; used ANSYS-APDL to model and simulate uterine synchronization.

National Institute of Technology Calicut *Jul 2004 – Jun 2008*

Bachelor of Technology in Mechanical Engineering *India*

RESEARCH INTERESTS

- Wall-modeled Large-Eddy Simulation (LES) of turbulent wall-bounded flows
- Non-equilibrium wall modeling, transitional flows, and relaminarization
- Data-driven and physics-informed modeling for CFD (Scientific Machine Learning)
- Reduced-order and surrogate modeling for engineering applications
- High-performance computing and large-scale numerical simulation
- Aerospace fluid dynamics and propulsion (compressible, aero-thermal flows)
- Urban environmental fluid mechanics and digital twins

WORK EXPERIENCE

Recognised Researcher – Barcelona Supercomputing Center *Aug 2024 – Jun 2025 | Barcelona, Spain*

- Generated 18+ high-fidelity urban flow simulations (≈14 TB each) spanning inlet flow directions from 0° to 340°, supporting AI-based digital twins for fast prediction of air pollutant distribution and odour impact in cities (APPWIND project)
- Resolved cross-platform data interoperability between BSC's in-house solver and the partner university's OpenFOAM environment: converted meshes to OpenFOAM format (ANSA) and exported velocity, pressure, and turbulent-viscosity fields as VTK (ParaView), troubleshooting node-numbering and partitioning issues to enable the partner's passive-scalar pollutant-dispersion simulations
- Contributed models and simulations within the vCity urban digital-twin platform as part of the platform's modeling team
- Mentored 3 graduate interns and master's students in numerical modeling, CFD workflows, and Python-based data analysis, providing technical guidance on in-house solvers and HPC environments.

First Stage Researcher – Barcelona Supercomputing Center *Sep 2018 – Jul 2024 | Barcelona, Spain*

- Conducted high-fidelity wall-modeled Large-Eddy Simulations of complex three-dimensional flows involving strong anisotropy, separation, and non-linear turbulent interactions, including NASA Junction flow, NASA Hump, and neutral atmospheric boundary layer flow over urban areas

- Developed a machine-learning-assisted wall model for LES that captured relaminarization (turbulent-to-laminar transition) — a capability physics-based wall models inherently lack — and outperformed the in-house equilibrium wall model in corners and recirculation regions
- Analysed large numerical datasets using Python-based scientific computing and data-analysis workflows
- Performed HPC-based CFD workflows using parallel computing environments, running simulations from under one million to 87 million elements
- Developed reproducible numerical-analysis and post-processing workflows for simulation data interpretation
- Authored a peer-reviewed conference paper as first author (published in Lecture Notes in Computer Science) and contributed to technical publications, conference papers, and presentations in CFD and HPC-related research areas
- Contributed high-fidelity LES data and data-driven wall-modeling research to the HifiTurb project's turbulence-model development
- Provided wall-modeling data and the data-driven wall model used in an Argonne National Laboratory study on XGBoost-based wall modeling in Nek5000; the resulting report cites the first-author conference paper

System Engineer – Indian Space Research Organization (ISRO) *Sep 2008 – Jul 2015 | Thiruvananthapuram, India*

- Served as System Engineer in the Air Breathing Propulsion Project, coordinating design, manufacturing, testing, and validation agencies
- Supported the SCRAMJET engine programme through its evolution from Technology Readiness Level (TRL) 3 to TRL 6, enabling a flight-testable prototype
- Coordinated ground-combustor testing, instrumentation, and validation activities for propulsion system components
- Covered the full component lifecycle — from engineering drawings (AutoCAD, Autodesk Inventor) through manufacturing, testing, qualification, and final integration
- Coordinated interdisciplinary teams operating under strict quality, reliability, and documentation standards
- Mentored graduate engineering trainees in engineering fundamentals and analytical problem-solving
- Participated in a vehicle characterisation launch campaign, flying dummy engine payloads on a sounding rocket to validate the carrier configuration ahead of the SCRAMJET flight test

PROJECTS

HifiTurb *Jul 2019 – Dec 2022*

- Development of high-fidelity LES/DNS datasets for innovative turbulence models; doctoral research on data-driven wall modeling for LES contributed to the project's turbulence-model development, resulting in a first-author paper in Lecture Notes in Computer Science.

APPWIND *Nov 2021 – Apr 2025*

- Development of high-resolution digital twins for fast prediction of air pollutant distribution and odour impact in cities, applying AI to CFD models; generated 18+ high-fidelity urban flow simulations (≈14 TB each) across inlet flow directions from 0° to 340° to build reduced-order-model datasets.

vCity *Sep 2023 – Jun 2025*

- Contributed urban flow simulations and POD-based dimensionality reduction to the vCity human-centric urban digital-twin platform; produced high-fidelity datasets (≈14 TB per simulation) and identified a parallel QR-factorization (LAPACK dorgqr) scaling constraint.

Air Breathing Propulsion Project *Sep 2008 – July 2015*

- Development of India's SCRAMJET engine, advancing the programme from TRL 3 to TRL 6 and participating in the vehicle-characterisation launch campaign ahead of the flight test.

PUBLICATIONS & TECHNICAL CONTRIBUTIONS

- Radhakrishnan, S., Calafell, J., Miró, A., Font, B., & Lehmkuhl, O. (2024). Data-driven wall modeling for LES involving non-equilibrium boundary layer effects. International Journal of Numerical Methods for Heat & Fluid Flow (Q1 journal), 34(8), 3166-3202.
- Radhakrishnan, S., Gyamfi, L. A., Miró, A., Font, B., Calafell, J., & Lehmkuhl, O. (2021). A data-driven wall-shear stress model for LES using gradient boosted decision trees. In International Conference on High Performance Computing, Lecture Notes in Computer Science (pp. 105-121). Cham: Springer International Publishing.
- Owen, H., Radhakrishnan, S., Gomez, S., & Lehmkuhl, O. (2019). A Finite-element Viscosity Based Non-slip Wall Modelling Strategy for the Simulation of Industrial Turbulent Flows. ERCOFTAC Bulletin, 120. https://doi.org/10.57651/b120.11

CONFERENCE PRESENTATIONS

International Supercomputing Conference – Online *2021*

Barcelona Supercomputing Center PhD Symposium – Barcelona, Spain *2021*

ParCFD Symposium – Alba, Italy *2022*

2nd Spanish Fluid Mechanics Conference – Barcelona, Spain *2023*

SKILLS

Numerical Modeling & Scientific Computing: Computational Fluid Dynamics (CFD), Large Eddy Simulation (LES), Turbulence Modeling and Wall Modeling, Reduced Order and Surrogate Modeling, Data-driven and AI/ML-assisted Modeling, Scientific workflow automation and post-processing, CFD data analysis and visualization, Verification and Validation (V&V), GMSH, ANSA, Finite Element Method (FEM), Finite Volume Method (FVM)
Programming & Automation: Python (NumPy, SciPy, Pandas, Matplotlib), Fortran, C/C++, Linux-based scientific computing workflows, Scientific scripting and data processing, Matlab, TensorFlow, scikit-learn
High-Performance Computing: HPC environments and parallel workflows, Large-scale simulation data processing, Numerical workflow optimisation
Engineering & Simulation Software: ANSYS (APDL & Mechanical), Finite element-based in-house code development, ParaView (VTK post-processing and visualization

MENTORING & COLLABORATION

Volunteer Science Outreach Lecturer

World Space Week (United Nations) — ISRO / Vikram Sarabhai Space Center (VSSC) *Oct 2010 – Oct 2014 | Kerala, India*

- Delivered outreach lectures to school students in rural areas on rocket science, satellites, and their benefits to humanity
- Participated in the United Nations' World Space Week celebrations, coordinated in India by ISRO and locally by Vikram Sarabhai Space Center (VSSC)
- Communicated complex aerospace topics to non-technical audiences, fostering early interest in science and space exploration

CERTIFICATIONS

Deep Learning Specialization – DeepLearning.AI *Jul 2021*

5-course specialization covering neural network architectures (CNNs, RNNs, LSTMs, Transformers) and training strategies (Dropout, BatchNorm, Xavier/He initialization), applied with Python and TensorFlow to real-world tasks including speech recognition, machine translation, and NLP.

Fundamentals of Parallelism on Intel Architecture – Intel *Aug 2021*

Covers the multiple forms of parallelism on modern Intel architecture processors — data parallelism with vector instructions, task parallelism with threads, distributed-memory parallelism with message passing, and containers — applicable to HPC, datacenter, and AI workloads.

LANGUAGES

English: Fluent
Malayalam: Native
Spanish: Basic
Tamil: Fluent
Hindi: Basic